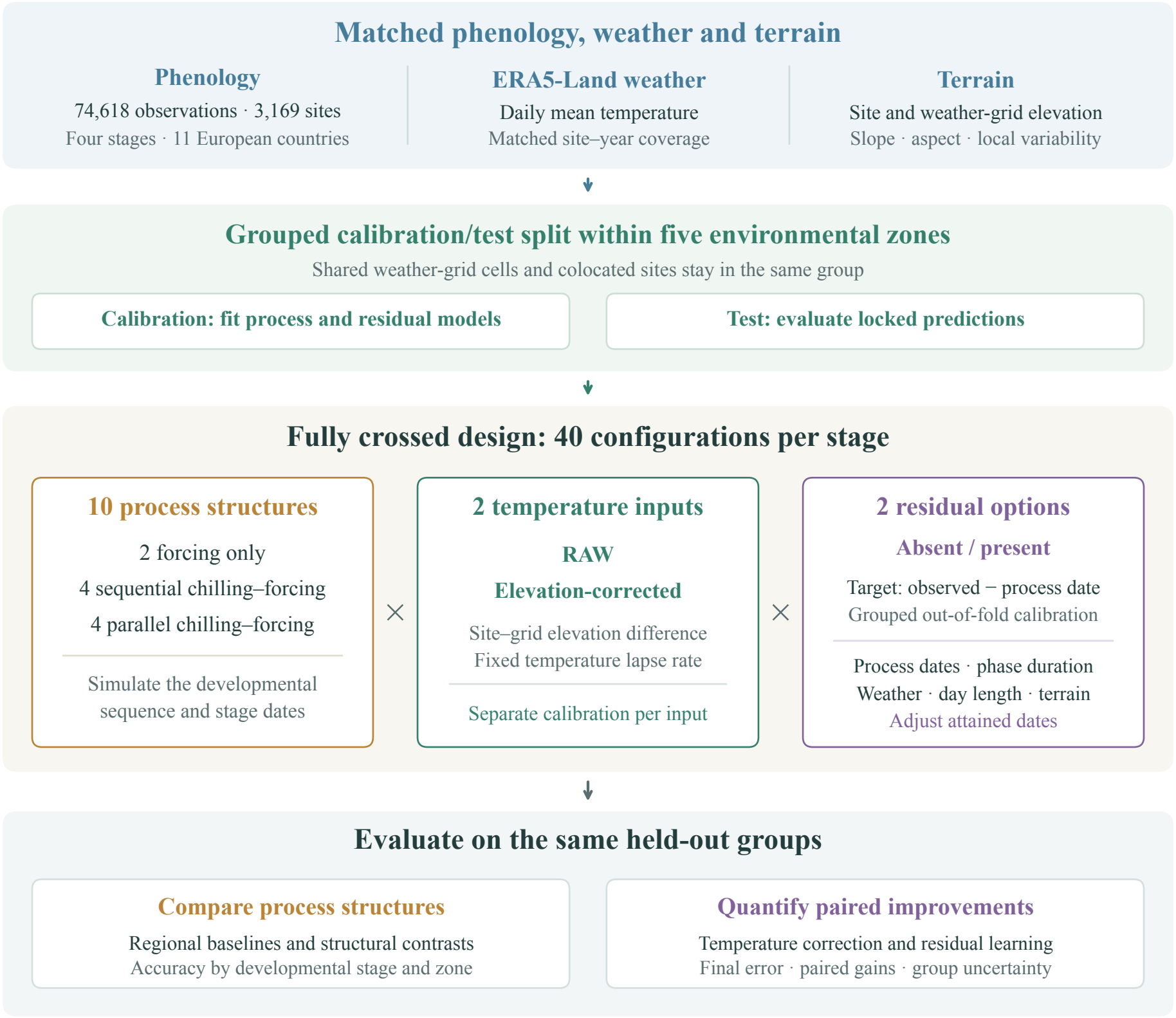

Matched phenology, weather and terrain
Phenology
74,618 observations · 3,169 sites
Four stages · 11 European countries
ERA5-Land weather
Daily mean temperature
Matched site–year coverage
Terrain
Site and weather-grid elevation
Slope · aspect · local variability
Grouped calibration/test split within five environmental zones
Shared weather-grid cells and colocated sites stay in the same group
Calibration: fit process and residual models
Test: evaluate locked predictions
Fully crossed design: 40 configurations per stage
10 process structures
2 forcing only
4 sequential chilling–forcing
4 parallel chilling–forcing
Simulate the developmental sequence and stage dates
×
2 temperature inputs
RAW
Elevation-corrected
Site–grid elevation difference
Fixed temperature lapse rate
Separate calibration per input
×
2 residual options
Absent / present
Target: observed − process date
Grouped out-of-fold calibration
Process dates · phase duration
Weather · day length · terrain
Adjust attained dates
Evaluate on the same held-out groups
Compare process structures
Regional baselines and structural contrasts
Accuracy by developmental stage and zone
Quantify paired improvements
Temperature correction and residual learning
Final error · paired gains · group uncertainty